

```
K4 = graphs.CompleteGraph(4)
K4.show( )
K4.is_planar( )
```

Upon executing the code above, a non-planar drawing of K_4 is displayed, as shown in Figure 6. However, since the message 'True' appears, it indicates that K_4 can indeed be drawn as a planar graph. Figure 7 shows a planar drawing of K_4 (courtesy of Wikipedia). I encourage you to read the Wikipedia article on 'Planar Graph' to learn more about this topic.

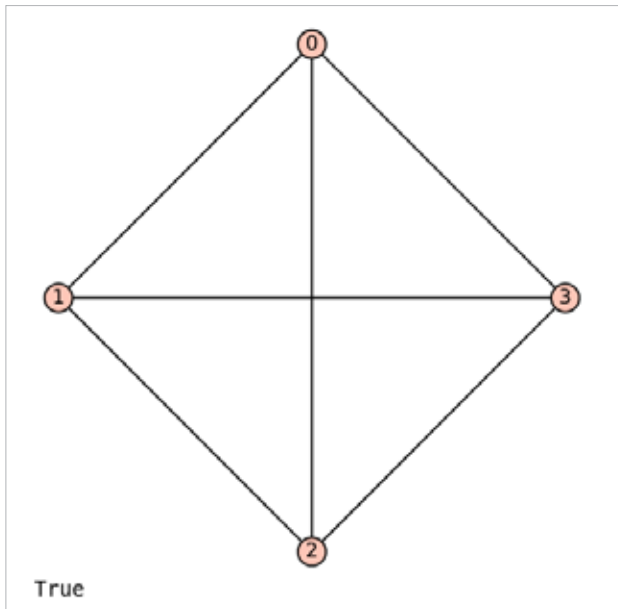


Figure 6: A non-planar drawing of K_4

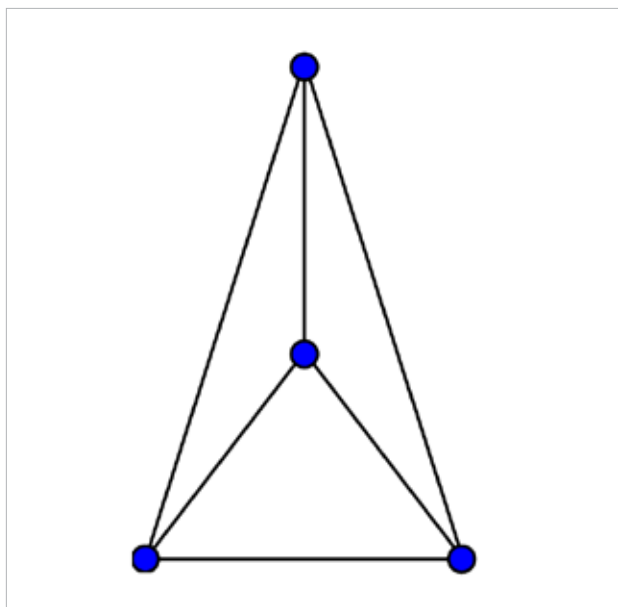


Figure 7: A planar drawing of K_4 (courtesy of Wikipedia)

Now, let us examine a straightforward example of a non-planar graph: the complete graph K_5 . The code below displays the complete graph K_5 and checks whether it is planar.

```
K5 = graphs.CompleteGraph(5)
K5.show( )
K5.is_planar( )
```

Upon executing the code above, a (non-planar) drawing of K_5 is displayed, as shown in Figure 8. Please take some time to verify that it is impossible to draw K_5 without edges crossing, which confirms that it is non-planar.

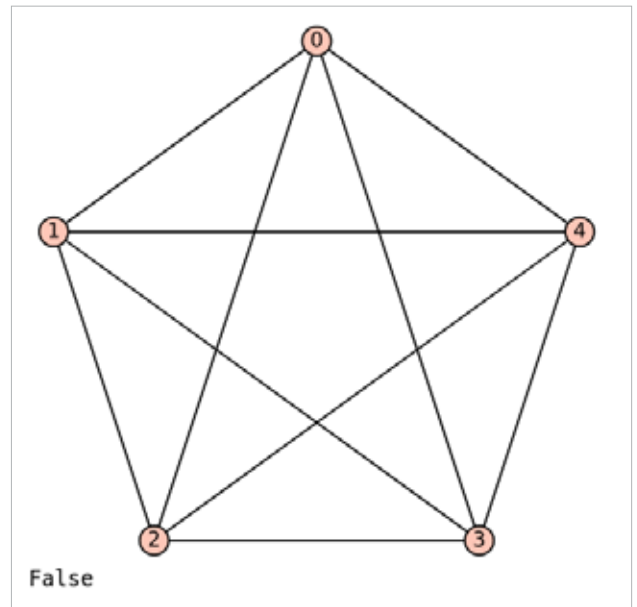


Figure 8: The complete graph K_5

Now, it is time to wrap up our discussion. We have been exploring graph theory for a while now, and while we could continue delving into it and still only scratch the surface, what we have covered should provide you with a strong foundation to begin your journey into graph theory and its applications in computer science and AI. Although we will occasionally revisit graph theory, our focus will shift to cybersecurity, another hot topic in computer science, starting with the next article. In the upcoming articles, we will explore how various concepts in cybersecurity can be implemented and tested using SageMath. **END** 🐧

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